

Sustainable Logistics

MSc2, Spring Semester 2026

Mikele Gajda
HEC Lausanne



Project Guidelines

Goal of the project

- The goal of the project is to make you **identify and define** some important aspects that may be improved in a logistics system of your choice using **optimization techniques and algorithms**. From a managerial point of view, the aim is to quantitatively show that these aspects play an important role in the ultimate efficiency of a company, as well as on its environmental impact.
- More particularly, you will consider Vehicle Routing Problem(s), Facility Location Problem(s), Packing Problem(s), or any other optimization model of your choice; solve it (them), and discuss the obtained results from a practical point of view.
- The aim is to focus on a situation that reflects managerial choices and **changes that could be made in a practical situation**, such that your work could act as a decision-aid tool and give insights about the relevance of the proposed choices and changes, as well as give new ideas to make the company's operations more sustainable and efficient.

Expected Output

- Description of the practical situation under study:
 - Practical **relevance**, practical examples.
 - Describe in detail the considered situation and set-up.
- When possible, provide with the **mathematical modelling** (objective function and constraints) of the considered problem, and describe the taken hypothesis and assumptions.
- Describe the data collected or generated for the different input parameters.
- Implement the programming set-up (with Google OR-Tools) to solve your problem, **provide the code**, and solve your problem(s).
- Discuss the obtained results:
 - Give **graphical representations** of some of the obtained solutions.
 - When appropriate, perform sensitivity analysis regarding the variation of (some of) the input parameters.
 - When different set-ups are considered, **compare the obtained solutions**.
 - Give managerial insights and recommendations

Solving your problems

When necessary, consider a simplified version of the (sub)-problems under study, to make it possible to solve them using the tools introduced during the course and within a reasonable amount of computing time.

Data

- In order to increase the practical relevance of your project, you are **encouraged to search for and collect real data** about existing companies/organizations and their potential customers/members to be used as input parameters to your logistics optimization problem(s). This data might include:
 - Geographical locations to be visited.
 - Potential locations to build facilities.
 - Distances between potential warehouses.
 - Energy costs, CO2 emissions, salaries, vehicle capacities, etc.
 - Etc.
- If not possible to obtain real data, **generate data as realistically as possible**. Motivate the hypothesis that have been taken.
- Provide the sources used for the data collection and/or the reasoning for the data generation.

Recommendations

- Try to give a solid structure to **YOUR thinking process**, i.e., Define - Measure - Analyze - Improve.
- Focus on instances of reasonable size first, such that they can be solved with the tools introduced in the course. Instance size can potentially be increased in a second stage.
- For instances of smaller size, provide graphical representations of the obtained solutions to get an intuition of their structure, their correctness, and their potential drawback(s)/advantage(s).
- When comparing different set-ups, focus on the obtained solutions, and show the potential trade-offs that might appear between them.
- Consider more than one instance, in order to reflect different situations that might appear in practice (e.g., urban vs. suburban areas, clustered locations vs. non-clustered locations, etc.)

Example of Projects

- Warehouse location and vehicle routing for a manufacture of pasta in 8 regions in France.
- Warehouse location and vehicle routing for an online grocery delivery company in Berlin.
- Warehouse location and vehicle routing for a fashion retailer in Switzerland, using their demand data and the transport plan from their logistics subcontractor.
- Parcel delivery in the 9th and 10th arrondissement of Paris in France.
- Warehouse location and vehicle routing for the delivery of COVID-19 vaccines in the Vaud canton, using the effective demand in the various vaccination centers.
- Optimization of delivery routes and minimization of the number of drivers for a start-up active in meal delivery.
- Introducing a cable car for goods delivery in Lausanne.
- Knapsack problem for an Itinerary planning tool for Interrail trips.
- Blood sample collection in the Vevey area.
- Scheduling of school shuttles in the North of Lausanne.

Inter-islands logistics in French Polynesia

Sustainable Logistics 2025

Group I

**Isabella Donner
Loan Dupire
Gaia De Gaetani
Mona El Khattabi
Mikael Fehlmann**





SUSTAINABLE LOGISTICS

Transportation of fruits and vegetables in the
canton of Vaud

EDA

Dataset Overview:

- Analyzed 1,216 potential planning units across the canton of Vaud.
- Each unit has data on: estimated energy production, distances (to infrastructure, protected areas), topography, visibility, and more.

High Potential Areas:

- Plateau

Mid Potential Areas:

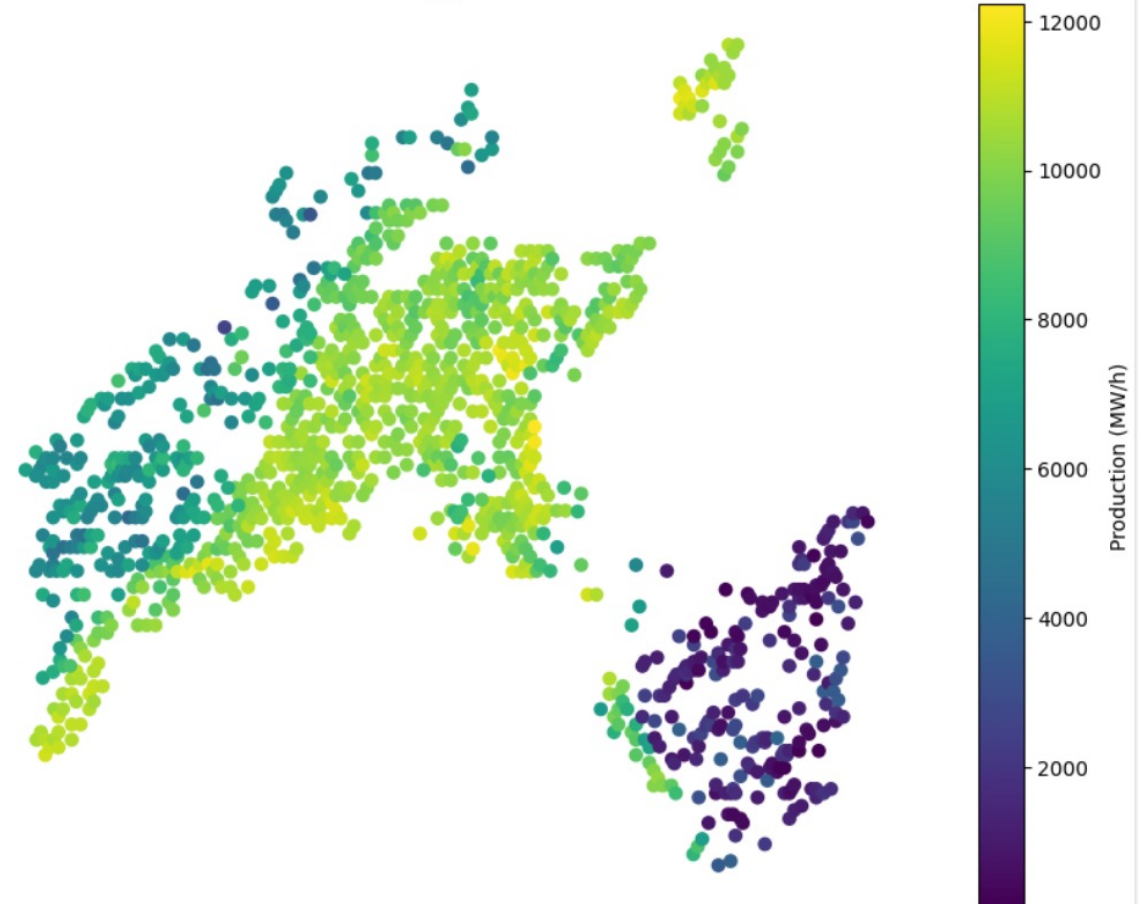
- Jura

Low Potential Areas

- Pre-alpes



Potential Wind Energy Production in Vaud



Deliverables

Synthetic written report, including an executive summary

To be sent by e-mail to mikele.gajda@unil.ch by Friday May 15th , 2025, 11:59pm.

Slides for the presentation

To be sent by e-mail to mikele.gajda@unil.ch by Monday May 18th , 2025, 8:00am.

Presentation to take place on **Monday May 18th , 2026** (see schedule on the Moodle, **ANT 4165**)

Presentation time: 15 minutes (+ 10 minutes Q&A)